

Ejercicio 2

$$h[m] = \delta[m] - \delta[m-4]$$

$$y[m] = x[m] * h[m] = x[m] - x[m-4]$$

a) $x[m] = \cos(\omega_0 \cdot m \cdot T_s)$

$$y[m] = \cos(\omega_0 \cdot m \cdot T_s) - \cos(\omega_0 (m-4) \cdot T_s)$$

Prop $\cos(A) - \cos(B) = -2 \sin\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$

• $A = \omega_0 \cdot m \cdot T_s$

• $B = \omega_0 m T_s - 4 \omega_0 T_s$

$$y[m] = -2 \sin(\omega_0 \cdot m \cdot T_s - 2 \omega_0 T_s) \cdot \sin(2 \omega_0 T_s)$$

Identidad $-\sin(\theta) = \cos(\theta + \pi/2)$

$$\theta = \omega_0 \cdot m \cdot T_s - 2 \cdot \omega_0 \cdot T_s$$

$$y[m] = \underbrace{\cos(\omega_0 \cdot m \cdot T_s - 2 \omega_0 \cdot T_s + \frac{\pi}{2})}_{\varphi} \cdot \underbrace{2 \sin(2 \omega_0 T_s)}_A$$

$$y[m] = 2 \cdot \sin(2 \omega_0 T_s) \cos(\omega_0 \cdot m \cdot T_s - 2 \omega_0 T_s + \frac{\pi}{2})$$

$$b) \quad y[n] = \left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{2}\right)^{n-4} u[n-4]$$

• $n < 0$

$$y[n] = \left(\frac{1}{2}\right)^n \cdot 0 - \left(\frac{1}{2}\right)^{n-4} \cdot 0 = 0$$

$$u[n] = 0$$

$$u[n-4] = 0$$

• $0 \leq n \leq 3$

$$u[n] = 1 \quad n \geq 0$$

$$u[n-4] = 0 \Rightarrow (\text{así } n=3 \Rightarrow 3-4 = -1 < 0)$$

$$y[n] = \left(\frac{1}{2}\right)^n \cdot 1 - \left(\frac{1}{2}\right)^{n-4} \cdot 0 = \left(\frac{1}{2}\right)^n$$

• $n \geq 4$

$$u[n] = 1 \Rightarrow n \geq 0$$

$$u[n-4] = 1 \Rightarrow n \geq 4 \Rightarrow n-4 \geq 0$$

$$y[n] = \left(\frac{1}{2}\right)^n \cdot 1 - \left(\frac{1}{2}\right)^n \cdot \left(\frac{1}{2}\right)^{-4} = \left(\frac{1}{2}\right)^n - \left(\frac{1}{2}\right)^n \cdot 16$$

$$y[n] = \left(\frac{1}{2}\right)^n - 16 \left(\frac{1}{2}\right)^n = \cancel{\left(\frac{1}{2}\right)^n} - 15 \left(\frac{1}{2}\right)^n$$

$$y[n] = -15 \left(\frac{1}{2}\right)^n$$

$$c) \quad x[n] = u[n+1] - u[n+2]$$

$$\bullet u[n+1] \Rightarrow 1 \text{ para } n = -1$$

$$\bullet u[n+2] \Rightarrow 1 \text{ para } n = 2$$

$$\bullet n < -1 : 0 - 0 = 0$$

$$\bullet n = 2 : 1 - 1 = 0$$

$$\bullet n = -1 : 1 - 0 = 1$$

$$\bullet n > 2 : 1 - 1 = 0$$

$$\bullet n = 0 : 1 - 0 = 1$$

$$\bullet n = 1 : 1 - 0 = 1$$

$$x[n] = \begin{cases} 1 & n = -1, 0, 1 \\ 0 & \text{en otro caso} \end{cases} \Rightarrow \text{la señal existe en sólo 3 instantes}$$

$$x[n] = \delta[n+1] + \delta[n] + \delta[n-1] \quad (\text{Seno de delta})$$

Reemplazo n por $n-4$, desplazando 4 posiciones a la derecha

$$x[n-4] = \delta[(n-4)+1] + \delta[n-4] + \delta[(n-4)-1]$$

$$x[n-4] = \delta[n-3] + \delta[n-4] + \delta[n-5]$$

$$x[n] = \begin{cases} 1 & \text{si } n = 3, 4, 5 \\ 0 & \text{en otro caso} \end{cases}$$

$$y[n] = x[n] - x[n-4]$$

$$\bullet n = -1 : 1 - 0 = 1$$

$$\bullet n = 3 : 0 - 1 = -1$$

$$\bullet n = 0 : 1 - 0 = 1$$

$$\bullet n = 4 : 0 - 1 = -1$$

$$\bullet n = 1 : 1 - 0 = 1$$

$$\bullet n = 5 : 0 - 1 = -1$$

$$\bullet n = 2 : 0 - 0 = 0$$

$$y[n] = \delta[n+1] + \delta[n] + \delta[n-1] - \delta[n-3] - \delta[n-4] - \delta[n-5]$$

$$y[n] = \{1, 1, 1, 0, -1, -1, -1\}$$

Bonus: DFT por 8 puntos

a) ~~señal~~

$$y[n] = \underbrace{2 \operatorname{sen}(2\omega_0 T_s)}_A \cos(\omega_0 n T_s - \underbrace{2\omega_0 T_s + \pi/2}_\varphi)$$

$$\bullet \omega_0 T_s = \frac{2\pi}{8} k_0$$

$$y[n] = A \cdot \frac{e^{j(\omega_0 n T_s + \varphi)} + e^{-j(\omega_0 n T_s + \varphi)}}{2}$$

$$y[n] = \frac{A}{2} e^{j\varphi} e^{j\frac{2\pi}{8} k_0 n} + \frac{A}{2} e^{-j\varphi} e^{-j\frac{2\pi}{8} k_0 n}$$

$$Y[k] = \sum_{n=0}^7 y[n] \cdot e^{-j\frac{2\pi}{8} k n}$$

$$Y[k] = \frac{A}{2} e^{j\varphi} \sum_{n=0}^7 e^{-j\frac{2\pi}{8} (k-k_0) n} + \frac{A}{2} e^{-j\varphi} \sum_{n=0}^7 e^{-j\frac{2\pi}{8} (k+k_0) n}$$

$$Y[k] = \frac{A}{2} e^{j\varphi} 8 \delta[k-k_0] + \frac{A}{2} e^{-j\varphi} 8 \delta[k-8+k_0]$$

$$Y[k] = 4A (e^{j\varphi} \delta[k-k_0] + e^{-j\varphi} \delta[k-8+k_0])$$

$$\bullet A = 2 \operatorname{sen}\left(\frac{\pi}{2} k_0\right) \rightarrow \text{Si } k_0 \text{ es par} \Rightarrow A = 0$$

$$Y[k] = 0$$

$$\bullet \text{ Si } k_0 \text{ es impar} = \pm 1 \quad |A| = 2$$

$$e^{j\varphi} = e^{j(-\frac{\pi}{2} k_0 + \frac{\pi}{2})} = e^{j\frac{\pi}{2} (1-k_0)}$$

$$k_0 = 1 \Rightarrow e^{j0} = 1$$

$$k_0 = 3 \Rightarrow e^{-j\pi} = -1$$

$$A \cdot e^{j\varphi} = 2 \operatorname{sen}\left(\frac{\pi}{2} k_0\right) e^{j\frac{\pi}{2} (1-k_0)} = 2$$

$$Y[k] = \begin{cases} 0, & \text{si } k_0 \text{ es par} \\ 8(\delta[k-k_0] + \delta[k-8+k_0]), & \text{si } k_0 \text{ es impar} \end{cases}$$

$$b) X[k] = \sum_{m=0}^7 x[m] e^{-j \frac{2\pi}{8} km}$$

$$X[k] = \sum_{m=0}^7 \left(\frac{1}{2}\right)^m e^{-j \frac{\pi}{4} km}$$

$$X[k] = \sum_{m=0}^7 \left(\frac{1}{2} e^{-j \frac{\pi}{4} k}\right)^m \quad \text{serie geometrica}$$

$$X[k] = \frac{1 - \left(\frac{1}{2} e^{-j \frac{\pi}{4} k}\right)^8}{1 - \frac{1}{2} e^{-j \frac{\pi}{4} k}} = \frac{1 - \frac{1}{256} e^{-j 2\pi k}}{1 - \frac{1}{2} e^{-j \frac{\pi}{4} k}}$$

$$e^{-j 2\pi k} = \cos(\underbrace{2k\pi}_1) - j \sin(\underbrace{2k\pi}_0) = 1$$

$$X[k] = \frac{1 - \frac{1}{256} \cdot 1}{1 - \frac{1}{2} e^{-j \frac{\pi}{4} k}} = \frac{\frac{255}{256}}{1 - \frac{1}{2} e^{-j \frac{\pi}{4} k}}$$

$$H[k] = \sum_{m=0}^7 h[m] e^{-j \frac{2\pi}{8} km}$$

$$H[k] = 1 \cdot e^{-j \frac{2\pi}{8} k(0)} - 1 \cdot e^{-j \frac{2\pi}{8} k(4)} = 1 - e^{-j \pi k}$$

Per Euler $e^{-j \pi k} = \cos(\pi k) - j \sin(\pi k) = (-1)^k$

$$H[k] = 1 - (-1)^k \rightarrow k \text{ pari} \Rightarrow H[k] = 0$$

$$\quad \quad \quad \downarrow$$

$$k \text{ dispari} \Rightarrow H[k] = 2$$

$$Y[k] = X[k] \cdot H[k] = \begin{cases} 0 & k \text{ pari} \\ \frac{255}{256} \cdot 2 & k \text{ dispari} \end{cases}$$

$$= \frac{255}{128} \cdot 2, \quad k \text{ dispari}$$

$$c) \quad x[n] = u[n+3] - u[n-2] = \text{entonces } u[n] - u[n-1]$$

~~pero el periodo es 8 puntos y desplazamiento periodo $m=8$ $n=7$~~
~~entonces $x[n] = u[n] - u[n-1]$~~

$x[n]$, me pide 8 puntos con desplazamiento periodo $m=8$ $n=7$

$$x[n] = \{1, 1, 0, 0, 0, 0, 0, 1\} = x[m] + x[m+8k]$$

$$x[-1] = x[-1+8] = x[7] = 1$$

$$X[k] = x[0] + x[1] e^{-j \frac{2\pi}{8} k} + x[7] e^{-j \frac{2\pi}{8} k}$$

$$X[k] = 1 + e^{-j \frac{2\pi}{8} k} + e^{-j \frac{\pi}{4} k} = 1 + 2 \cos\left(\frac{\pi}{4} k\right)$$

$$Y[k] = X[k] H[k] \rightarrow k \text{ par } Y[k] = 0$$

$$k \text{ impar } Y[k] = X[k] \cdot 2$$

$$Y[k] = \left[1 + 2 \cos\left(\frac{\pi}{4} k\right) \right] \cdot 2 \quad \text{para } k \text{ par}$$

$$Y[k] = \begin{cases} 0 & k \text{ par } 0, 2, 4, 6 \\ 2 + 4 \cos\left(\frac{\pi}{4} k\right) & k \text{ impar } 1, 3, 5, 7 \end{cases}$$